

3.2 Semantic Search as a Geometry Problem: Cosine Similarity

Semantic Search

- Semantic search is a geometry problem:
 - Embed your documents, embed your query, find what's closest.
- Cosine similarity** measures the angle between two vectors:

$$\text{cosine similarity} = S_C(A, B) := \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

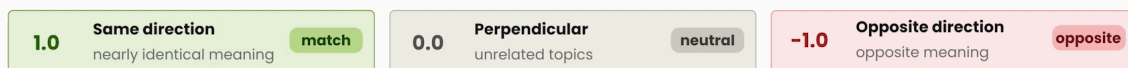


Figure 8. Semantic search as geometry. Embed documents and query, then rank by cosine similarity, where 1.0 means same direction, 0.0 means perpendicular, and -1.0 means opposite.

✦ **Abstract.** Semantic search reduces to nearest-neighbor search: embed the corpus once, embed each query, and return the closest vectors. **Cosine similarity** measures the angle between vectors and ignores magnitude. At scale, exact search gives way to **approximate nearest-neighbor (ANN)** indexes that trade a little recall for orders-of-magnitude speed.

🔍 Deep Technical Deconstruction

The formula.

$$S_C(A, B) = \cos(\theta) = (\mathbf{A} \cdot \mathbf{B}) / (\|\mathbf{A}\| \|\mathbf{B}\|) = \sum_i A_i B_i / (\sqrt{\sum A_i^2} \cdot \sqrt{\sum B_i^2})$$

The three reference points on the slide.

cos θ	Geometry	Slide interpretation	Practical reality
1.0	Same direction ($\theta = 0^\circ$)	Nearly identical meaning	Near-duplicates and paraphrases score 0.85–0.95
0.0	Perpendicular ($\theta = 90^\circ$)	Unrelated topics	With anisotropic models, "unrelated" often scores 0.2–0.5
-1.0	Opposite ($\theta = 180^\circ$)	Opposite meaning	Rarely observed; antonyms usually score <i>high</i>

IMPORTANT

The -1.0 "opposite meaning" case is a geometric idealization. In trained text-embedding models, antonyms such as "increase the budget" and "decrease the budget" share topic and context, so they typically have **high positive** similarity. Embeddings encode *relatedness*, not *logical agreement*. Use cross-encoders or LLM verification when polarity matters.

Cosine, dot product, and Euclidean distance. For unit-normalized vectors ($\|\mathbf{A}\| = \|\mathbf{B}\| = 1$):

$$\mathbf{A} \cdot \mathbf{B} = \cos \theta \text{ and } \|\mathbf{A} - \mathbf{B}\|^2 = 2 - 2 \cos \theta$$